Washington University in St. Louis
SCHOOL OF MEDICINE

Eric C. Leuthardt, MD, MBA
Washington University in St. Louis
Department of Neurological Surgery
Box 8057, 660 South Euclid Ave
Saint Louis, MO 63110
leuthardte@wustl.edu
March 5th, 2026

Re: Comprehensive Profiles of Human Cingulate Gyrus Effective Networks

To the editorial board at Imaging Neuroscience,

It is with great pleasure that we are submitting our manuscript titled "**Effective Network Profiles of the Human Cingulate Gyrus**" for consideration for publication in *Imaging Neuroscience*.

This work combines macro-level effective connectivity profiling of the human cingulate gyrus with methodological advances in quantifying cortico-cortical evoked potentials (CCEPs, the voltage responses recorded across the cortex during direct invasive electrical stimulation of target regions). We believe that both of these contributions align with your journal's scope of systems-level brain organization and neurophysiological methodology.

The networks of the human cingulum are studied most often due to their complexity, relevance to cognition and behavior, and role in various disease states. Recent clinical advances have enabled interventional methods to define and characterize such networks, specifically through direct electrical stimulation of brain regions and subsequent analysis of evoked responses across the cortex. Using stereoelectroencephalography (SEEG)- mediated single-pulse electrical stimulation, we mapped the effective networks of the anterior, mid, and posterior cingulate gyrus. This manuscript makes three contributions that we believe are relevant to this journal's readership. First, we provide subregion-resolved effective connectivity profiles of the cingulate, characterizing an anterior-posterior connectivity gradient, shared cortical hub regions, differential subnetwork organization, and hemispheric asymmetries in connectivity. Second, we introduce a CCEP coherence metric that quantifies waveform morphology across stimulation trials, detecting connections missed by conventional magnitude-based measures. Third, using random forest classification, we determined the predictive strength of effective connectivity features, demonstrating that conventional classification methods, such as N1/N2 peaks, carry the least predictive value. It is often asserted that the current standards for evaluating effective connectivity are not sufficient, but to our knowledge, no other work has yet performed direct comparisons between measures of effective connectivity and their utility.

We believe this paper offers a generalizable, data-driven framework that improves the practical standards for quantifying and comparing effective networks. The cingulate is one of the most studied brain regions, and as such, quantitative models of its effective networks will provide support to any research focused on elucidating its function, physiology, or role in pathology. **Our paper represents an important advance in understanding the connections of the human cingulate and the quantitative methods used to model its networks.** For these reasons, we believe that this work is impactful and suited for publication in *Imaging Neuroscience.*

We thank you for considering our manuscript for publication and look forward to your decision.

Yours sincerely,

Eric C. Leuthardt, MD, MBA
Professor of Neurological Surgery, Neuroscience, Biomedical Engineering, Mechanical Engineering, and Materials Science
Chief, Division of Neurotechnology
(on behalf of the authors)